3 Apportionment

3.1 Jefferson's Method

Definition 3.1.1

- Find the Standard divisor
- Find each states lower quota
- If the lower quotas do not add up to the number of seats available, then adjust the divisor, and go back to step 2.

Example 3.1.2 Let's redo the first example from last time, this time with Jefferson's method.

County	Population
Kent	162,310
New Castle	538,479
Sussex	197,145
Total	1,052,567

Example 3.1.3 Use Jefferson's method to apportion the 75 seats of Rhode Island's House of Representatives amoung it's five counties.

County	Population
Brisol	49,875
Kent	166,158
Newport	82,888
Providence	626,667
Washington	126,979
Total	1,052,567

Example 3.1.4 Three peaople invest in a treasure dive, each investing the amount listed below. The dive results in 36 gold coins. Apportion those coins to the investors using Jefferson's method.

Alice	\$7,600
Ben:	\$5,900
Calos:	\$1,400

3.2 Apportionment of Legislative Districts

In most states, rather than apportioning each county a number of representatives, districts are drawn so that each legislator represents a district. Therefore, the apportionment process happens by drawing the districts themselves. Unfortunately, the process of redistricting is typically done by the legislature itself, so we often see **gerrymandering**.

Definition 3.2.1 **Gerrymandering** is when districts are drawn based on the political affiliation of the constituents to the advantage of those drawing the boundary.

Example 3.2.2 Suppose that based on previous voting patters, you predict that the city in which you live (which is a perfect rectangle!) will vote according to the picture below:

There is an election coming up, and you are a legislator in the yellow party. Luckily, you are in charge of redistricting the city into 5 districts before the next upcoming election. Can you find 5 districts so that

- Each district must contain at least one yellow square and at least one blue square, and
- 3 of the 5 districts will vote yellow (there are more yellow squares than blue ones in that district)?